

Shubham Shaw

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🐙 github.com/ShubhamShaw2006 | 🌐 shubhamshaw2006.github.io/shubham-spidey-portfolio

OBJECTIVE

Third-year CS student with hands-on experience across the data stack — from SQL-based pipelines and BI dashboards to applied machine learning. Seeking Data Analyst, Data Engineering, or AI/ML internship roles to build reliable, insight-driven data systems.

EDUCATION

Institute of Engineering and Management
B.Tech in Computer Science & Business Systems

Kolkata, India
Expected: 2028

EXPERIENCE

IEM Centre of Excellence for Cloud Computing & IoT

Intern

Kolkata, India

October 2024 – July 2025

- Built Python data analysis pipelines using Seaborn and Matplotlib to visualize and compare experimental results across routing scenarios, enabling clearer interpretation of simulation outputs
- Designed and simulated drone routing models using ONE Simulator to benchmark multiple DTN routing protocols, analyzing performance trade-offs across delivery probability, latency, and overhead metrics
- Prototyped distributed inference workflows on NVIDIA AGX Thor edge hardware and implemented IoT automation in Node-Red with Arduino Nano; co-authored technical documentation on the findings

PROJECTS

E-Commerce Customer Retention & Delivery Performance Analysis *Team Project* 🐙 [GitHub](#)

- Collaborated on analyzing ~110K orders from a 9-table relational e-commerce dataset using PostgreSQL, resolving one-to-many join issues across payments and reviews via aggregation CTEs
- Identified that 97% of customers never make a repeat purchase and quantified \$243,560 in recoverable revenue from 988 lapsed customers through RFM segmentation in Python; found late deliveries nearly halve satisfaction (2.3/5 vs. 4.2/5)
- Designed a 4-page interactive Power BI dashboard (Executive Overview, Customer Segments, Delivery Performance, Recommendations) connected live to PostgreSQL

Tech Layoffs Analysis: Risk Patterns 2020–2026 🐙 [GitHub](#) 📊 [Live Dashboard](#)

- Designed a PostgreSQL schema and built a Python ETL pipeline unifying 4,500+ layoff events (Kaggle) with live Yahoo Finance and U.S. BLS API data, handling mismatched date formats and inconsistent API responses
- Found early-stage companies cut 48% of workforce on layoff vs. 15% for late-stage, and firms with under \$10M raised cut 61% of staff vs. 18% for those with over \$500M raised
- Built composite risk-scoring SQL logic and published an interactive Tableau Public dashboard visualizing risk patterns by industry, funding stage, and time

Hypervisor-Level Static Malware Detection Using Random Forest Ensemble

- Built a static malware detection model on the BODMAS dataset using extracted PE header features, achieving 96–99% accuracy, precision, recall, and F1-score on the held-out test set
- Implemented an optimized Random Forest ensemble in Scikit-learn with feature importance analysis to identify the most discriminative PE attributes; developed reproducible training and evaluation pipelines

Legal-Vani: AI-Powered Legal Assistance Platform 🐙 [GitHub](#) 📺 [Live Demo](#)

- Built a multilingual, voice-first legal aid platform (Next.js, Tailwind CSS) supporting 12 Indian languages, enabling users to file complaints by speaking in their native language
- Integrated Whisper AI and Groq LLMs to convert speech into structured legal advice with automated PDF generation; architected secure Supabase (PostgreSQL) backend with JWT auth and Overpass API geolocation

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, MATLAB

Databases & Data Engineering: PostgreSQL, Schema Design, ETL Pipelines, REST APIs, CTEs, Window Functions

ML & Data Science: Scikit-learn, Random Forest, Feature Engineering, Model Evaluation, RFM Analysis, Pandas, NumPy

BI & Visualization: Power BI, Tableau, Seaborn, Matplotlib

Tools & Platforms: Git, Linux, Supabase, Node-Red, ONE Simulator, NVIDIA AGX Thor, Arduino Nano, Figma

Web & AI Frameworks: Next.js, Tailwind CSS, Whisper AI, Groq LLMs, Vercel

CERTIFICATIONS

- 🔥 Crash Course on Python – Google
- 🔥 Data Structures and Algorithms using Java – IIT Kharagpur (NPTEL)
- 🔥 IoT Essentials for Professionals: Smart Tech & Automation

LEADERSHIP

- 👤 Secretary, IETE ISF – IEM: Coordinate the chapter's flagship Pan-India Hackathon and manage its LinkedIn presence
- 👤 Student Coordinator, IEM IEDC: Support institute-wide innovation and entrepreneurship initiatives